

THE UNIVERSITY OF HONG KONG

**SCHOOL OF
COMPUTING & DATA SCIENCE
DEPARTMENT OF COMPUTER
SCIENCE**

**COMP7503 Multimedia Technologies
Deadline: 6 Nov 2026, 23:55**

Preamble:

In this assignment, you are required to record two short audio clips of your own voice (or of tones that you generate and capture with a microphone), then analyse them with spectral analysis. The first learning goal is to read a sound from both the time-domain waveform and the spectrogram. The second learning goal is to design the two clips so that you can verify two classic psychoacoustic effects used in perceptual audio coding: temporal masking and frequency (simultaneous) masking.

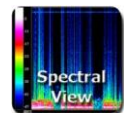
You must create the stimuli yourself, submit the recordings together with screenshots, and explain what the plots and your listening test show.

Recommended tools

Android Platform:

Recommended: Spectral Audio Analyzer

<https://play.google.com/store/apps/details?id=radonsoft.net.spectralview&hl=en>



iOS Platform:

Recommended: SpectrumView

<https://itunes.apple.com/us/app/spectrumview/id472662922?mt=8>



After you install these apps, you should grant the permission for the App to access the microphone on your mobile device.

Desktop (optional but useful for controlled tones): Audacity (free), or any DAW that can display both waveform and spectrogram. An online tone generator on a second device may be used to produce masker and probe tones.

Recording requirements (both clips)

Record in a quiet environment. Use a sampling rate of at least 8 kHz if the tool allows it. Keep each submitted clip short (about 3–8 seconds). Do not apply noise reduction, equalisation, or compression that would hide the masking effect. Name the files clearly, for example Audio1_temporal.wav and Audio2_frequency.wav. WAV or lossless format is preferred; if you must submit compressed audio, state the codec and bitrate.

What to record

Audio Clip 1 — Temporal masking. Design a sequence that contains a loud, short masker followed by a quieter, shorter probe. Record at least two conditions in the same clip (or as two takes inside Clip 1): (i) a very short gap after the masker (about 10–30 ms), and (ii) a clearly longer gap (about 150–300 ms). A practical realisation is a loud spoken syllable or clap (masker) followed by a quiet syllable or a quiet tone (probe). Forward masking is the main effect to demonstrate: a strong sound can make a weaker sound that comes immediately afterwards harder to hear, even though the weaker sound is still visible on the waveform or spectrogram. You may also comment on backward masking if you place a quiet probe immediately before a loud masker.

Audio Clip 2 — Frequency (simultaneous) masking. Design a sequence in which a loud masker and a quieter probe occupy overlapping time. Record at least two conditions: (i) the probe frequency is close to the masker (for example a loud 1 kHz tone plus a quiet tone near 1.1–1.2 kHz, or a loud sustained vowel plus a quiet nearby whistle), and (ii) the same probe level but at a frequency well away from the masker (for example near 4 kHz). Also record the quiet probe alone at the same level as a control. A loud low-frequency sound can mask a weaker nearby frequency more than a distant one; this is simultaneous frequency masking and is related to critical bands.

Questions:

1. Spectrum analysis of your two recordings. For each of Audio Clip 1 and Audio Clip 2, submit: (a) the time-domain waveform; (b) the spectrogram (frequency vs time, with a colour or grey scale for energy). On the plots, annotate the masker, the probe, and the time gaps or frequency locations you intended. Briefly explain how to read the displays: what the waveform shows about amplitude over time, and what the spectrogram shows about how energy is distributed across frequency. Point out at least the following in each spectrogram: regions of strong energy, weaker energy, and (if present) broadband vs narrowband structure. [8 marks]

2. Verify the temporal masking effect using Audio Clip 1. Report a simple listening test: is the probe clearly audible after the long gap, but reduced or inaudible after the short gap, at the same probe level? Use the waveform and spectrogram to show that the probe is physically present in both conditions (energy appears at the expected time). Then explain why hearing the probe can still fail when it closely follows a louder sound. Relate your observation to temporal masking as used in perceptual audio coding (for example, why a coder may allocate fewer bits to a weak sound that falls inside a forward-masking window). State the approximate gap durations you used. [6 marks]

3. Verify the frequency masking effect using Audio Clip 2. Report the listening test for the nearby-frequency probe versus the distant-frequency probe, and versus the probe played alone. Use the spectrogram to show that the probe energy is present in the masked condition (a ridge or patch at the probe frequency) even if it is hard to hear. Explain simultaneous (frequency) masking and why a masker is more effective against a nearby frequency than against a remote one. Relate the result to critical bands and to frequency masking in perceptual audio coding. [6 marks]

Submission

Submit a single PDF report plus the two audio files (or a zip containing the report and the two audio files) via the course submission system before the deadline. The report must include screenshots of waveforms and spectrograms with your annotations, short written answers to Questions 1–3, and a statement of the tools, sampling rate, and how each clip was produced. Name your files with your University Number.

Academic conduct

The recordings must be made by you. Spectrograms copied from the web, or from another student, are not acceptable. You may discuss concepts with classmates, but the clips, plots, and write-up must be your own work.

=== End ===